AI-Assisted Federal Survey Harmonization

Cross-Survey Integration Analysis of 47 Census Bureau Demographic Surveys

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Chapter 1

AI-Assisted Federal Survey Harmonization

DRAFT — These results are preliminary and have not been independently validated or peer reviewed.

Full methodology, data, and code: github.com/brockwebb/federal-survey-concept-mapper

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Chapter 2

Introduction

The Census Bureau conducts a broad portfolio of surveys covering demographics, economics, housing, health, education, and more. This research examines 47 of those survey instruments — a substantial cross-section of the Bureau's demographic survey programs — containing approximately 7,000 questions in total. These 47 instruments do not represent the full scope of Census Bureau data collection, but they cover much of the demographic survey landscape and provide a meaningful test bed for cross-survey analysis.

Each of these surveys was designed for a specific analytical purpose. The Current Population Survey tracks labor force dynamics. The American Community Survey provides community-level demographic and economic profiles. The Consumer Expenditure Survey measures household spending. The National Health Interview Survey captures health status and access to care. And so on, across dozens of specialized instruments fielded on different schedules, to different populations, with different reference periods.

These surveys were built independently, and for good reason: different research questions demand different measurement approaches. But that independence comes at a cost. Where surveys ask similar questions about similar topics, those overlapping questions represent an untapped resource. A shared question about household income on two different surveys can serve as a bridge variable — a connection point that allows analysts to link findings across surveys, increasing the analytical value of each collection without adding respondent burden.

No comprehensive map of these connections has existed. Building one manually would require a research team to compare thousands of questions across dozens of instruments, a task that takes months and exceeds practical limits not because of complexity, but because of scale. No individual, however expert, holds 7,000 questions in working memory at once.

This research used AI-assisted methods to build that map at scale, with built-in safeguards for transparency, traceability, and bias mitigation — including independent models from multiple vendors and structured arbitration for disagreements. The complete analysis was conducted in weeks at a total cost under $100 in computing resources. The methodology is described briefly in context throughout this report, with architectural diagrams in Appendix A and full details in the companion Methodology and TEVV documents.

This report presents the results: a scored inventory of harmonization opportunities organized around the American Community Survey as a central reference point. For two survey pairs examined in detail — CPS and FoodAPS, each compared against ACS — the inventory identifies which questions have direct counterparts, which require statistical adjustment to align, and which measure fundamentally different constructs. These findings are intended to support survey managers in identifying where cross-survey integration adds value and where instruments can be better aligned.

Chapter 3

Classification

Before two surveys can be compared, their questions need a common vocabulary. A question about "total household income before taxes" on the CPS and a question about "total income from all sources" on the ACS may measure the same thing, but without a shared label, the connection is invisible at scale. Classification provides that shared label.

Two independent large language models — OpenAI GPT-5-mini and Anthropic Claude Haiku 4.5 — classified each of the 6,954 analyzable questions into the Census Bureau's Survey Explorer taxonomy of five topics and 152 subtopics. Using models from different vendors serves as a built-in reliability check: agreement between architecturally distinct systems is stronger evidence than agreement within a single model family.

The two models agreed on topic-level classification for 89.2% of questions (Cohen's = 0.839), indicating near-perfect agreement by standard interpretation scales. Subtopic-level agreement was 69.7% (= 0.687), reflecting the finer granularity and greater ambiguity at that level. Both exceed thresholds for substantial agreement, confirming that the classification is not an artifact of any single model's tendencies. Where the two models disagreed, a higher-capability model (Claude Sonnet 4.5) arbitrated, applying the same taxonomy with access to both prior judgments.

The classified questions distribute unevenly across topics:

Topic	Count	Share
Economic	~2,980	42.8%
Social	~2,467	35.5%
Housing	~967	13.9%
Demographic	~369	5.3%
Government	~167	2.4%

Economic and Social questions together account for more than three-quarters of all content. This concentration is not surprising — it reflects the Census Bureau's core demographic survey mission — but it has consequences for harmonization. Surveys that both emphasize economic measurement will share more subtopic territory than surveys spanning different domains, which shapes where cross-survey connections are most likely to emerge.

The subtopic classifications are what make cross-survey comparison possible. When a CPS question and an ACS question both receive the subtopic label "Employment Status," they become candidates for detailed pairwise evaluation. The next chapter identifies which surveys share the most subtopic territory with ACS and selects the pairs for that evaluation.

The complete dual-model classification cost approximately $15 in API fees and finished in roughly two hours of wall-clock time.

Chapter 4

Survey Overlap and Selection

With 47 survey instruments classified into a common taxonomy, the next step is identifying which surveys are measuring in the same conceptual neighborhoods. Evaluating every possible pair — over 1,000 combinations — is neither practical nor necessary. Instead, the analysis uses the American Community Survey as an anchor: the reference point against which other surveys are compared.

ACS is not the oldest federal household survey — that distinction belongs to the Current Population Survey, which has run continuously since 1948. But ACS covers the broadest range of topics across demographics, economics, housing, and social characteristics, and its results serve as control totals and denominators throughout the federal statistical system. That topical breadth makes it the most useful anchor: if a question on another survey has a conceptual counterpart anywhere, ACS is the most likely place to find it.

Using the subtopic classifications from the previous chapter, the analysis identifies which surveys share the most subtopic territory with ACS. Five surveys form ACS's closest "family" and constitute the focused scope of this research:

Survey	Shared Subtopic Intersections	Subtopics Covered	Top Overlap Areas
SIPP	577	38	Health Insurance, Employment Status, Household
AHS	460	34	Structure, Utilities, Plumbing/Kitchen
CE	283	30	Health Insurance, Housing Costs, Household
CPS	181	25	Employment Status, Earnings, Hours/Weeks
FoodAPS	123	23	Food Stamps (SNAP), Household, School Enrollment

The "shared subtopic intersections" column counts question-by-subtopic overlaps, not unique questions — a survey with many questions in a shared subtopic scores higher. This captures both breadth of overlap and depth of coverage within shared topics. The top overlap areas hint at where harmonization candidates are most likely to concentrate.

An important distinction: concept overlap does not imply question overlap. Two surveys that both classify questions under "Employment Status" are measuring in the same neighborhood, but their questions may differ in wording, reference period, response format, or target population. Sharing a subtopic means the surveys are trying to measure or account for similar things — it narrows the search space for potential harmonization but does not guarantee it. Whether specific question pairs are actually compatible is the work of the next chapter.

From this family, the analysis selects CPS and FoodAPS to prove out the pairwise methodology. The choice is pragmatic: these two surveys produce the fewest question pairs, making them the least expensive to run through the full multi-model evaluation pipeline. Validating the approach on the cheapest pairs first — before committing resources to SIPP (577 intersections) or AHS (460) — is straightforward cost optimization. CPS and FoodAPS also happen to represent different survey profiles: CPS is large and broadly demographic, FoodAPS is small and specialized. Together they contribute 380 unique source questions (240 from CPS, 140 from FoodAPS), a sufficient sample to validate the approach before scaling up.

Chapter 5

Method

The previous chapter identified five surveys in ACS's family and selected CPS and FoodAPS — the two producing the fewest question pairs — to validate the methodology at minimal cost. This chapter describes how those pairs were evaluated.

Questions sharing the same subtopic classification between a source survey and ACS are paired for direct comparison. Rather than relying on assumptions about which questions should match, the analysis takes a naive combinatorial approach: every source question is paired with every ACS question in its subtopic. This produces 1,030 pairs for CPS and 568 for FoodAPS — 1,598 pairs total, drawn from 380 unique source questions. The approach is deliberately agnostic so that the method generalizes to any survey pair without requiring prior knowledge of which questions correspond.

Most of these pairs are expected to fail. A CPS question about employment status might be paired with an ACS question about industry simply because both fall under the same economic subtopic. Within any subtopic, typically only one or two ACS questions will be genuinely comparable to a given source question — the rest are conceptual neighbors but not measurement equivalents. The expected exceptions are standard demographic items (name, age, race, sex) that appear on virtually every survey in similar form. The goal is not to achieve a high match rate across all pairs; it is to surface the highest-probability matches as a review list for subject-matter experts.

Each pair is evaluated on two dimensions. First, feasibility: can these questions serve as bridge variables? Three tiers apply, adapted from the DataSHaPER compatibility framework used in international survey harmonization programs (Fortier et al. (2011)):

Code	Meaning	What it takes
F1	Direct recode	Same construct, compatible formats. Collapse categories or convert units.
F2	Statistical adjustment	Same construct, different measurement. Requires a documented transformation.
F3	Incompatible	Different constructs, or barriers no statistical method can bridge.

Second, for pairs rated F3, a barrier code identifies the specific obstacle. Six codes capture the ways questions can fail to align, drawn from the survey harmonization literature (Wolf et al. (2016); Slomczynski & Tomescu-Dubrow (2018)):

Code	Barrier	Example
CC	Construct/Concept	"Do you want full-time work?" vs. "Did you work last week?"
TC	Temporal	"Hours worked last week" vs. "usual hours per year"
RS	Response Scale	Open dollar amount vs. bracketed income categories
PC	Population/Coverage	Adults 18+ vs. persons 15+
MC	Mode/Context	Self-administered web vs. interviewer telephone
PM	Processing/Metadata	SOC 2010 codes vs. SOC 2018 without crosswalk

Full definitions, subtypes, and literature provenance are in Appendix B.

Three independent AI models — one each from OpenAI, Anthropic, and Google — rate every pair on both dimensions. Using models from competing vendors is a deliberate trust architecture: no single vendor's training data, alignment approach, or reasoning tendencies can dominate the results. This is the methodological equivalent of requiring independent expert raters from different institutions.

At the initial rating stage, the three models achieved moderate agreement (Fleiss' = 0.611 for barrier classification, 0.537 for feasibility). Where raters disagree, three independent arbitrator models — each vendor's more capable model — review the dispute with access to all prior judgments and render a verdict. Arbitration substantially improved agreement:

Metric	Rater Stage ( )	Post-Arbitration ( )
Feasibility	0.537	**0.843**
Primary barrier	0.611	0.796
Binary (harmonizable/not)	—	**0.896**

Post-arbitration feasibility agreement (= 0.843) falls in the "almost perfect" range. Both quality gates passed. The pattern — moderate initial agreement converging to near-perfect after structured arbitration — validates that the task is well-defined and that independent models can reach consensus when given a protocol for resolving ambiguity.

The full methodology, including prompt templates, model configurations, behavioral analysis, and the NIST AI Risk Management Framework crosswalk, is documented in the companion TEVV and Methodology documents. Appendix C provides the NIST trustworthiness mapping.

Chapter 6

The previous chapter described a method for evaluating whether survey questions measuring similar concepts can actually be harmonized: pair every source question with every ACS question sharing its subtopic, then have three independent AI models rate each pair's feasibility, with arbitration resolving disagreements. This chapter reports what that process found.

The CPS contains 211 questions and FoodAPS contains 462 questions across four questionnaire instruments. Not all of these entered the analysis. Only questions whose subtopic classifications overlapped with ACS were eligible for pairing — 164 unique CPS questions (78% of the survey) and 123 unique FoodAPS questions (27% of the survey). The difference reflects each survey's conceptual distance from ACS: CPS shares broad labor force and demographic territory, while FoodAPS is predominantly a specialized food acquisition instrument whose questions on shopping frequency, food storage, and meal preparation have no ACS counterpart. These raw question counts also overstate what any single respondent encounters, since both surveys use branching logic that routes respondents through different question paths based on their circumstances.

Each eligible question was paired with every ACS question sharing its subtopic, producing 1,030 CPS–ACS pairs and 568 FoodAPS–ACS pairs. Most of these pairs were expected to fail — a question about weekly earnings and a question about commute mode might share an "Economic" subtopic but measure entirely different things. The purpose of the broad search was to ensure no viable match was missed. The following table summarizes the funnel from total survey questions through to harmonization results:

	CPS	FoodAPS
Total survey questions	211	462
Questions with ACS concept overlap	164 (78%)	123 (27%)
Question pairs evaluated	1,030	568
Questions with harmonization paths	**93 (56.7% of paired)**	**61 (49.6% of paired)**
As share of total survey	**44.1%**	**13.2%**

Across both surveys, 154 unique questions are candidate bridge variables for expert review. These results vary by subject matter:

Survey	Topic	Questions	Harmonizable	Rate
CPS	Demographic	29	23	79.3%
CPS	Economic	101	58	57.4%
CPS	Social	33	12	36.4%
FoodAPS	Demographic	12	10	83.3%
FoodAPS	Economic	59	32	54.2%
FoodAPS	Social	42	13	31.0%
FoodAPS	Housing	10	6	60.0%

Demographic questions harmonize at the highest rates — roughly 80% across both surveys. This is expected: federal surveys use relatively standardized approaches to collecting age, sex, race, and similar basic characteristics. The more substantive finding is in the non-demographic categories. Among CPS economic questions, the largest pools of bridge variable candidates are earnings (19 of 23 questions), employment status (20 of 34), and hours worked (12 of 18). Among FoodAPS economic questions, SNAP participation (11 of 22 questions), employment status (8 of 9), and school enrollment (8 of 11) contribute the most. Full subtopic breakdowns appear in Appendix C.

The next chapter examines what these results mean in practice — which harmonization paths are straightforward, which require methodological investment, and what the barrier patterns reveal about the feasibility of broader cross-survey integration.

Chapter 7

The previous chapter established that 93 CPS questions and 61 FoodAPS questions have at least one harmonization path to ACS — 154 candidate bridge variables in total. This chapter examines what those paths look like in practice and what the pattern of failures reveals about the broader feasibility of cross-survey integration.

Not all harmonization paths require the same effort. The 154 candidates span a range from mechanical recoding to substantive methodological work. Some questions — primarily demographic items like age, sex, race, and citizenship — were identified as high-confidence matches before the barrier assessment even began: both classification models independently agreed they were easily harmonizable. These represent the floor of what cross-survey integration can achieve with minimal investment. The remaining candidates emerged from the barrier assessment pipeline, where three independent AI models evaluated each question pair's feasibility with arbitration resolving disagreements. Questions rated F1 require only direct recoding — mapping one survey's response categories onto another's. Questions rated F2 measure the same underlying concept but differ in ways that require statistical adjustment: reference period differences (weekly versus annual earnings), response granularity (detailed occupation codes versus broad categories), or population framing (slightly different eligibility criteria). These F2 differences are real, but they are the kind of differences that established survey methodology can address through calibration, imputation, or bridging studies.

The topic-level pattern from the previous chapter showed demographic questions harmonizing at roughly 80%, economic questions around 55%, and social questions around 33%. The subtopic detail reveals where the substantive opportunity concentrates:

Survey	Subtopic	Harmonizable	Total	Rate
CPS	Earnings	19	23	82.6%
CPS	Employment Status	20	34	58.8%
CPS	Hours/Week	12	18	66.7%
FoodAPS	SNAP Participation	11	22	50.0%
FoodAPS	Employment Status	8	9	88.9%
FoodAPS	School Enrollment	8	11	72.7%

These are core analytic variables that researchers routinely wish they could combine across surveys. CPS earnings questions bridging to ACS would enable labor market analyses that link detailed weekly earnings data with the rich demographic and geographic detail ACS provides. FoodAPS SNAP questions bridging to ACS would connect food assistance participation patterns with community-level characteristics. These are not hypothetical use cases — they are analyses that currently require either restricted-use linked data or imperfect proxy measures.

Social questions harmonize at the lowest rates (36% CPS, 31% FoodAPS), reflecting genuine variation in how surveys operationalize concepts like disability assessment, veteran status verification, and educational attainment. These differences often reflect deliberate design choices tied to each survey's legislative mandate, making them harder to bridge without compromising measurement intent.

For questions that cannot be harmonized, the barrier classification explains why:

Barrier	Description	Share of Incompatible Pairs
CC — Construct mismatch	Questions measure fundamentally different things	~87%
TC — Temporal	Reference period differences	~7%
RS — Response scale	Different answer formats	~4%
PC — Population	Different target populations	~1%
MC — Mode/context	Different collection methods	~1%
PM — Precision	Different measurement granularity	<1%

Construct mismatch dominates because the pairing strategy was deliberately broad — every source question was matched against every ACS question sharing its subtopic. Most of these pairs are not genuine matches, and the methodology correctly identified them as such. The more actionable finding is the roughly 13% of barriers that involve potentially addressable differences. Reference period misalignment (7%) and response scale differences (4%) represent concrete design opportunities: a survey redesign that aligns reporting windows or harmonizes answer categories could convert some incompatible questions to harmonizable ones, expanding the bridge variable inventory without requiring new data collection.

The pipeline also demonstrates a practical division of labor between AI and human expertise. Of the 1,598 question pairs evaluated, the vast majority required no expert attention — the AI models agreed on clear mismatches or clear matches. The 154 candidate bridge variables are the deliverable that goes to subject-matter experts for validation. Rather than asking experts to evaluate thousands of pairs, the methodology reduces the review burden to a curated set where expert judgment adds the most value: confirming that statistically viable harmonization paths are also substantively meaningful, and that the barrier classifications accurately reflect the nature of each mismatch.

The next chapter considers what these findings imply for the broader federal survey ecosystem and where this methodology goes from here.

Chapter 8

Limitations and Next Steps

This analysis evaluates two survey pairs — CPS–ACS and FoodAPS–ACS — out of 46 possible pairings with ACS as anchor, and many more across the full 47-survey topology. The 164 CPS questions and 123 FoodAPS questions assessed here represent those with Census topic overlap with ACS. Of these, 154 questions (93 CPS, 61 FoodAPS) have viable harmonization paths, yielding rates of 56.7% and 49.6% respectively. The consistency of these rates across two different survey types is encouraging, but broader coverage is needed to confirm the pattern holds. The remaining survey pairs — including high-value connections like CPS–SIPP, NHIS–AHS, and cross-domain pairings between economic and social surveys — await similar assessment. The combinatorial scale is substantial: with 47 surveys, there are over 1,000 possible pairwise comparisons before considering multi-hop paths.

The harmonization assessments are AI-generated using multi-model rating and arbitration. Post-arbitration agreement on feasibility ratings is high (= 0.843), but the results have not been validated against independent expert judgment. The methodology intentionally pre-filters easily harmonizable pairs (demographic basics like age, sex, citizenship) and routes only barrier cases through detailed coding, maximizing resource efficiency while maintaining assessment quality. Expert validation should focus on the 275 barrier-coded questions to confirm AI judgment aligns with domain expertise, particularly for questions rated at the F1/F2 boundary where statistical adjustment requirements are most sensitive to context.

The current pairing strategy joins every source question with every ACS question sharing a Census subtopic classification. This simple approach ensures coverage but produces many unrelated pairs, inflating the pair-level denominator. More targeted matching — using semantic embeddings or concept similarity — could improve efficiency while maintaining recall. The question-level aggregation applied here mitigates this limitation by focusing on the stakeholder-relevant metric: how many source questions have at least one viable match, regardless of the total number of pairs generated.

ACS serves as anchor given its comprehensive coverage of Census demographic topics, but it is not the only reference point. Some survey pairs may have stronger harmonization potential with each other than either has with ACS. Stage 4 addresses this through multi-survey network analysis, discovering integration paths across the full topology including multi-hop connections where Survey A harmonizes with B, B with C, enabling indirect linkage. The total analysis cost for this proof of concept was under $100, demonstrating that AI-based feasibility assessment can operate at a scale infeasible for manual expert review. The methodology is designed to extend beyond Census Bureau surveys to the broader federal statistical system, enabling cross-agency integration assessment at negligible marginal cost per survey pair.

Part I

Appendices

Architecture and Data Flow

Pipeline architecture diagrams for each analysis stage are in development. The analysis proceeds through five stages: topic classification, multi-model rating, arbitration, findings rollup, and deliverable generation.

Diagrams will be added in a subsequent revision.

Harmonization Barrier Taxonomy

This appendix defines the barrier codes and feasibility classifications used throughout this report. The taxonomy draws on established survey harmonization frameworks, adapted for application to federal demographic surveys. Each code is grounded in the survey methodology literature and mapped to its primary source below.

Framework Provenance

The classification schemes used in this analysis are not novel constructs. They are adaptations of frameworks developed for and deployed in large-scale, operational survey harmonization programs worldwide.

The feasibility tiers (F1/F2/F3) derive from the DataSHaPER/Maelstrom Research framework (Fortier et al., 2011; Fortier et al., 2017), which was developed through collaboration with over 100 investigators across 15 countries and tested across 53 large epidemiological studies encompassing 6.9 million participants. The Maelstrom guidelines have since been adopted by the BioSHaRE-EU consortium (Biobank Standardisation and Harmonisation for Research Excellence in the European Union), the Canadian Partnership for Tomorrow Project, the BBMRI-LPC (Biobanking and Biomolecular Resources Research Infrastructure — Large Prospective Cohorts), the InterConnect global diabetes research initiative, and the Swiss Population Health Network Cohort Consortium, among others. The Maelstrom Research catalogue hosts metadata from over 370 individual studies across multiple international networks, a number that continues to grow as new consortia adopt the framework.

The barrier codes (TC, CC, PC, RS, MC, PM) synthesize terminology from several complementary frameworks. Wolf et al. (2016), published in the SAGE Handbook of Survey Methodology, codifies harmonization barrier categories drawn from decades of practice in cross-national survey programs including the International Social Survey Programme (ISSP; 45 member nations since 1984), the European Social Survey (ESS; 31 countries), and the Eurobarometer (European Commission, continuous since 1974). Slomczynski & Tomescu-Dubrow (2018) contributes the Survey Data Recycling (SDR) framework, which harmonized data across 23 international survey programs covering political attitudes, social capital, and well-being. Saris & Gallhofer (2014) provides the response scale quality dimensions used in psychometric and survey design practice internationally. Oleksiyenko et al. (2018) addresses processing error identification across the same cross-national survey ecosystem.

The comprehensive state-of-the-art reference is Tomescu-Dubrow et al. (2024), which consolidates ex-ante and ex-post harmonization methodologies across these programs.

In short: every code in this taxonomy maps to a concept that has been operationalized, tested, and refined across programs spanning dozens of countries, millions of respondents, and decades of continuous operation. We adapted these frameworks for application to federal demographic surveys within a single national statistical system — a narrower scope than the cross-national programs where they originated, but one that faces the same fundamental harmonization barriers.

Feasibility Classification

The feasibility tier assigned to each question pair reflects the practical effort required to use one survey's question as a bridge variable for another. This three-tier scheme adapts the DataSHaPER compatibility framework (Fortier et al., 2011; Fortier et al., 2017).

Code	Name	Definition	Practical Implication
F1	Direct Recode	Questions measure the same construct with compatible response formats. Harmonization requires only deterministic recoding, category collapsing, or unit conversion.	High-quality bridge variable. Usable with minimal documentation.
F2	Statistical Adjustment	Questions measure the same underlying construct but differ in measurement approach — reference period, response scale, or level of detail. Harmonization requires a documented statistical transformation with stated assumptions.	Usable bridge variable with known limitations. Requires methodological transparency.

F3	Incompatible	Questions measure fundamentally different constructs or have barriers that no statistical method can adequately address.	Not a bridge variable. The barrier code explains the specific obstacle.

In the DataSHaPER literature, Fortier et al. (2011) found that 38% of variable-study pairs across 53 studies could be harmonized, with "essential" demographic variables achieving 62% compatibility. Our analysis finds broadly comparable rates for federal demographic surveys (see **?@sec-results**).

Barrier Codes

When a question pair is classified F3 (incompatible), one of six barrier codes identifies the nature of the obstacle. These codes synthesize multiple established frameworks: Wolf et al. (2016) on question-level harmonization barriers, Slomczynski & Tomescu-Dubrow (2018) on methodological variability across survey programs, Saris & Gallhofer (2014) on response scale characteristics, and Oleksiyenko et al. (2018) on processing errors. Table 8.2 summarizes the taxonomy with literature provenance.

Table 8.2: Harmonization barrier codes with literature provenance.

Code	Name	Definition	Primary Source	Example from Data
TC	Temporal	Reference period or timing differences prevent meaningful comparison	Wolf et al. (2016); Fortier et al. (2011)	"Hours worked last week" vs. "usual hours worked per year"
CC	Construct	Concept definition or operationalization differences — the questions measure different things despite shared topic area	Wolf et al. (2016); Fortier et al. (2011)	"Do you want to work full-time?" (intention) vs. "Did you work last week?" (behavior)
PC	Population/Coverage	Target population, sampling frame, or eligibility criteria differ between surveys	Slomczynski & Tomescu-Dubrow (2018)	Adults 18+ vs. persons 15+; civilian noninstitutional vs. all residents
RS	Response Scale	Scale type, category structure, or response format differences prevent direct comparison	Saris & Gallhofer (2014)	Open-ended dollar amount vs. income brackets with different boundaries
MC	Mode/Context	Interview mode or questionnaire context introduces systematic differences	Wolf et al. (2016)	Self-administered web survey vs. interviewer-administered telephone survey
PM	Processing/Metadata	Coding schemes, derived variable algorithms, or documentation gaps prevent assessment	Oleksiyenko et al. (2018)	SOC 2010 occupation codes vs. SOC 2018 codes without crosswalk

Table 8.3 maps each code to its corresponding terminology in the source literature, enabling readers familiar with those frameworks to locate our analysis within the established landscape.

Table 8.3: Literature crosswalk mapping taxonomy codes to established terminology.

Our Code	Wolf et al. (2016)	Fortier et al. (2011)	Slomczynski & Tomescu-Dubrow (2018)
TC	Reference period differences	Reference period	Temporal variability
CC	Concept differences	Inferential equivalence	Source variable properties
PC	—	Observational units	Sampling design
RS	Response format differences	—	—
MC	Questionnaire context	Mode of administration	Mode effects
PM	—	—	Data processing

Barrier Subtypes

Each barrier code includes subtypes that capture the specific nature of the difference. Subtypes are documented in the analysis data and available for expert review but are summarized here for reference.

TC — Temporal Constraints

Based on Wolf et al. (2016) "reference period differences" and Fortier et al. (2011) finding that reference period characteristics predict harmonization potential.

- **TC.1 Reference period length.** Different recall windows (e.g., 7-day vs. 12-month).
- **TC.2 Temporal framing.** Point-in-time vs. habitual vs. retrospective framing (e.g., "last week" vs. "usually").
- **TC.3 Calendar alignment.** Fixed calendar periods vs. rolling reference periods (e.g., "calendar year 2025" vs. "past 12 months").

CC — Construct Constraints

Based on Wolf et al. (2016) "concept differences" and Fortier et al. (2011) requirement of "inferential equivalence." CC is the dominant barrier category in our data.

- **CC.1 Concept definition.** Different meaning of core term (e.g., "employment" including or excluding unpaid family work).
- **CC.2 Operationalization.** Different behavioral indicators for nominally the same concept (e.g., 12-item vs. single-question food insecurity measurement).
- **CC.3 Boundary conditions.** Different thresholds or cutoffs (e.g., "full-time" defined as 35+ vs. 40+ hours).
- **CC.4 Scope inclusions.** Different components counted (e.g., "income" including or excluding capital gains).

PC — Population/Coverage Constraints

Based on Slomczynski & Tomescu-Dubrow (2018) treatment of "methodological variability between survey projects" including sampling design.

- **PC.1 Universe definition.** Target population differs (e.g., civilian noninstitutional vs. all residents).
- **PC.2 Frame exclusions.** Different sampling frame exclusions (e.g., group quarters, territories).
- **PC.3 Age bounds.** Different age eligibility (e.g., 15+ vs. 16+ vs. 18+).
- **PC.4 Geographic scope.** Different geographic coverage (e.g., 50 states vs. 50 states plus territories).

RS — Response Scale Constraints

Based on Saris & Gallhofer (2014) response scale quality dimensions.

- **RS.1 Scale type.** Fundamentally different response formats (e.g., binary vs. 5-point Likert).
- **RS.2 Category structure.** Different numbers or boundaries of categories within the same scale type.
- **RS.3 Anchoring/labels.** Different verbal anchors or category labels.
- **RS.4 Numeric vs. verbal.** Numeric scales vs. labeled categories (e.g., 0–10 slider vs. frequency labels).

MC — Mode/Context Constraints

Based on Wolf et al. (2016) "questionnaire context differences."

- **MC.1 Interview mode.** Different data collection modes (e.g., CATI vs. web self-administered).
- **MC.2 Question routing.** Different skip patterns affect who answers and in what context.
- **MC.3 Contextual priming.** Preceding questions affect interpretation.
- **MC.4 Proxy response.** Different rules for proxy vs. self-report.

PM — Processing/Metadata Constraints

Based on Oleksiyenko et al. (2018) identification of processing errors in cross-national surveys.

- **PM.1 Coding schemes.** Different classification systems (e.g., SOC 2010 vs. SOC 2018).
- **PM.2 Derived variables.** Different construction algorithms for composite measures.
- **PM.3 Documentation gaps.** Insufficient metadata to assess comparability.

Coding Hierarchy

When multiple barriers apply to a single pair, the primary barrier is assigned according to the following hierarchy, which reflects the degree to which each barrier type is fundamental vs. remediable:

1. **CC** (Construct) — most fundamental; questions measure different things
2. **TC** (Temporal) — affects what the measurement captures
3. **RS** (Response Scale) — limits mechanical comparability
4. **PC** (Population) — affects who is measured
5. **MC** (Mode/Context) — introduces systematic measurement effects
6. **PM** (Processing) — technical rather than conceptual

All applicable barrier codes are documented in the analysis data; the hierarchy determines which code is reported as primary.

References

All citations are listed in the report bibliography. The primary sources for this taxonomy are:

- Wolf et al. (2016) — core barrier categories for question-level harmonization
- Fortier et al. (2011) and Fortier et al. (2017) — DataSHaPER/Maelstrom compatibility framework and feasibility tiers

- Saris & Gallhofer (2014) — response scale quality dimensions
- Slomczynski & Tomescu-Dubrow (2018) — ex-post harmonization methodology and population/sampling variability
- Oleksiyenko et al. (2018) — processing error identification
- Tomescu-Dubrow et al. (2024) — comprehensive state-of-the-art reference

TEVV Framework

This appendix presents the Test, Evaluation, Verification, and Validation (TEVV) evidence supporting the AI-assisted methodology used in this research. The report claims that multiple AI models can classify survey question pairs with sufficient reliability to guide expert review and inform harmonization decisions. This appendix provides the evidence for that claim — and states honestly where the evidence is established, where it is partial, and where it requires further validation.

Approach

The TEVV framework maps the quality measures built into the analysis pipeline against the NIST AI Risk Management Framework (AI RMF 1.0) trustworthiness characteristics. This is not a compliance exercise — NIST AI RMF is a voluntary framework, and this research is not a deployed AI system. The crosswalk serves a different purpose: it demonstrates that the pipeline's quality measures address recognized dimensions of AI trustworthiness, and it identifies where gaps remain.

The key insight from related work in AI-mediated federal statistical data consultation is that domain quality frameworks (in our case, the survey harmonization literature) and AI trustworthiness frameworks (NIST AI RMF) address overlapping but distinct concerns. Standard survey harmonization practice does not account for AI-specific failure modes such as fabricated harmonization paths or confidently asserted equivalence where none exists. Conversely, the NIST framework does not address domain-specific concerns like construct equivalence or reference period alignment. A complete evaluation requires both.

Crosswalk: Pipeline Quality Measures × NIST AI RMF

Table 8.4 maps each pipeline quality measure to the NIST AI RMF characteristic it addresses, with the current evidence status.

Table 8.4: Pipeline quality measures mapped to NIST AI RMF 1.0 trustworthiness characteristics.

Pipeline Quality Measure	What It Demonstrates	NIST AI RMF Characteristic	Evidence Status
Multi-vendor rater agreement (3 independent models from different vendors classify each pair)	Independent replication: classifications are not artifacts of a single model's training data or biases	Valid & Reliable	Established. Cohen's and Fleiss' computed at rater stage.
Randomized presentation order (question A/B order randomized for each rater)	Order invariance: results are not artifacts of presentation sequence	Valid & Reliable	Established. Randomization documented in pipeline.
Multi-vendor arbitration (3 independent arbitrators from different vendors resolve disagreements)	Vendor independence at resolution stage: arbitration outcomes are not dominated by one vendor's behavior	Valid & Reliable; Accountable & Transparent	Established. Behavioral analysis shows distinct arbitrator patterns (synthesis rates, selection biases).
Structured arbitration protocol (arbitrators see all three rater outputs, must select or synthesize, must provide reasoning)	Decision traceability: every arbitrated outcome has a documented rationale	Accountable & Transparent; Explainable & Interpretable	Established. All arbitration records include reasoning field.
Public code, prompts, and data (full pipeline, all prompt templates, and all anonymized outputs are available)	Transparency and reproducibility: another researcher can inspect and replicate the entire analysis	Accountable & Transparent	Established. GitHub repository with complete pipeline.
Deterministic pipeline (same inputs, same random seeds, same outputs)	Reproducibility: results are not artifacts of non-deterministic inference	Valid & Reliable	Established. Pipeline uses fixed seeds; checkpoint/resume architecture.
Human SME review (domain experts review a sample of classified pairs to validate assignments)	Classification accuracy: the AI assigned the correct barrier codes and feasibility tiers	Valid & Reliable	By design, not yet executed. SME review protocol defined below.

Pipeline Quality Measure	What It Demonstrates	NIST AI RMF Characteristic	Evidence Status
AI-specific failure mode awareness (see **?@sec-ai-failure-mode**)	Fabrication detection: the AI did not invent harmonization paths that do not exist	Valid & Reliable	Partially addressed. Multi-vendor agreement provides indirect evidence; direct validation requires SME review.

NIST Characteristics Not Applicable

Three NIST AI RMF characteristics do not apply meaningfully to this research context:

- **Safe.** This is an offline classification task on public survey metadata. There is no deployment context where misclassification creates physical safety risk. The relevant harm — misguided harmonization decisions — is addressed through the SME review gate before any operational use.
- **Secure & Resilient.** The pipeline processes public metadata through commercial APIs. There are no adversarial attack surfaces relevant to the classification task itself. Standard API security practices apply but are not a research contribution.
- **Privacy-Enhanced.** All input data is public survey questionnaire text. No personally identifiable information is processed at any stage.

These are stated as not applicable rather than forced into the framework. The NIST AI RMF explicitly notes that not all characteristics apply to all AI systems.

Fair with Harmful Bias Managed

The relevant bias concern in this context is epistemic rather than demographic: do the AI models systematically favor or disfavor certain barrier codes, feasibility tiers, or survey pairs? The behavioral analysis of arbitrator patterns (see **?@sec-results**) provides direct evidence on this question. Different vendors exhibit measurably different synthesis rates and selection patterns, which is why multi-vendor design is a trustworthiness feature rather than a convenience. No single vendor's behavioral pattern determines the final classification.

The AI-Specific Failure Mode

Standard survey harmonization quality frameworks address whether harmonization decisions are methodologically sound — correct construct mapping, appropriate statistical adjustments, defensible feasibility assessments. They do not address an AI-specific failure mode: **confident fabrication of harmonization paths that do not exist.**

An AI model could, in principle, assert that two questions measure the same construct when they do not, or propose a statistical bridge where the underlying concepts are incommensurable. This is the "N/A cell" in the domain framework × NIST AI RMF crosswalk: the survey harmonization literature was not designed to address AI failure modes, and the absence of coverage is expected rather than a deficiency in those frameworks.

Multi-vendor agreement mitigates this risk indirectly — independent fabrication of the same incorrect harmonization path by three different vendors is unlikely. But indirect mitigation is not validation.

Direct validation of classification accuracy requires human domain experts who know these surveys to review a sample of classified pairs and confirm or reject the AI's assignments. This is why SME review is a design feature of the methodology, not an afterthought. The pipeline produces the candidate classifications and confidence-ranked expert review lists; human judgment determines whether those classifications are correct.

SME Review Protocol

The following protocol is designed for the planned expert validation stage:

Sample selection. Pairs are drawn from each triage quadrant to ensure coverage across the confidence spectrum. The two-axis triage framework (direction × stability) prioritizes review effort toward cases where it matters most: edge cases where models agree on direction but disagree on barrier specifics (high direction, low stability) and ambiguous cases where neither direction nor barrier assignment is clear (low direction, low stability). High-confidence pairs from both the consolidable and non-consolidable quadrants are also sampled to verify that AI confidence tracks accuracy.

Review format. For each sampled pair, the expert sees the two question texts, the assigned barrier code and feasibility tier, and the reasoning provided by the model. The expert is asked: (1) Is the barrier code correct? (2) Is the feasibility tier correct? (3) If not, what is the correct classification and why?

Agreement metric. Expert-AI agreement is computed per barrier code and feasibility tier. Systematic disagreement patterns (e.g., experts consistently upgrading F3 to F2 for a particular barrier type) are diagnostic and inform taxonomy refinement.

What constitutes success. The methodology does not require perfect AI-expert agreement. It requires that AI classifications are sufficiently accurate to direct expert attention efficiently — that the triage system puts the right pairs in front of the right reviewers. Residual disagreement between AI and expert judgment is expected and informative.

Summary

The pipeline provides strong evidence of process reliability: multiple independent models agree at measured rates, presentation order does not determine outcomes, arbitration follows a transparent protocol, and the entire analysis is reproducible and publicly inspectable.

What the pipeline does not provide — by design — is direct evidence of classification accuracy. That evidence requires human expert review, which the methodology is designed to produce as a next step. This is not a gap; it is the boundary between what AI-assisted methods can establish autonomously (process quality) and what requires human domain expertise (outcome quality). The TEVV framework makes this boundary explicit.

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